ECE 3355: Electronics

Exam 3

Complete the exam on your own, without the help of your notes, prior examples or solutions, your book, or any communication/interaction with others. You must write a complete solution that shows the relevant steps if you want full credit for the problem. You may use a calculator and a crib sheet is provided as part of this exam. **You will have 1 hr. 15 min. to finish the exam.**

Student's Name: ______________________________

Question	Points	Score
1	45	
2	55	
Total:	100	

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha}\right) e^{v_{BE}/V_T}$$

Note: For the pnp transistor, replace v_{BE} with v_{EB}.

$$i_C = \alpha i_E$$

$$i_C = \beta i_B$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$i_B = (1-\alpha)\, i_E = \frac{i_E}{\beta+1}$$

$$i_E = (\beta+1)\, i_B$$

$$\alpha = \frac{\beta}{\beta+1}$$

$V_T = \text{thermal voltage} = \dfrac{kT}{q} \simeq 25\,\text{mV}$ at room temperature

Summary of Table 7.3 (Small Signal Model Parameters)

Model Parameters in Terms of DC Bias Currents

$$g_m = \frac{I_C}{V_T} \quad r_e = \frac{V_T}{I_E} = \alpha\frac{V_T}{I_C} \quad r_\pi = \frac{V_T}{I_B} = \beta\frac{V_T}{I_C} \quad r_o = \frac{|V_A|}{I_C}$$

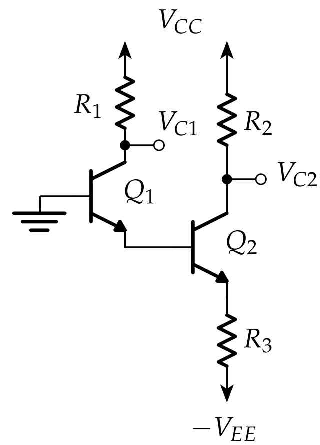
In Terms of g_m

$$r_e = \frac{\alpha}{g_m} \qquad r_\pi = \frac{\beta}{g_m}$$

In Terms of r_e

$$g_m = \frac{\alpha}{r_e} \qquad r_\pi = (\beta+1)\, r_e \qquad g_m + \frac{1}{r_\pi} = \frac{1}{r_e}$$

1. Consider the following circuit where $R_1 = 10\ \text{k}\Omega$, $R_3 = 300\ \Omega$, $V_{CC} = 14$ V, $-V_{EE} = -14$ V and β=99.

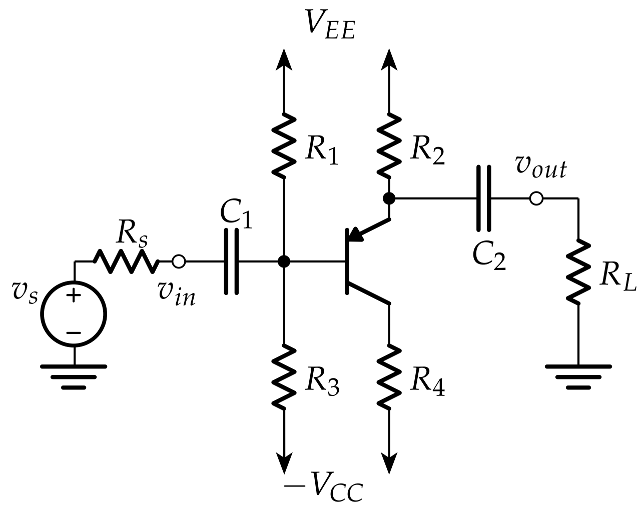


(a) (15 points) Find R_2 so that $V_{C2} = 1$ V. Is Q2 in an 'active' state?

(b) (15 points) Find V_{C1}. Is Q1 in an 'active' state?

(c) (15 points) What values of R_2 ensure that Q2 is in an 'active' state?

3. Use the following circuit to complete this problem.



(a) (5 points) Assuming that the transistor is in an active mode, sketch the small signal model for the circuit. *Hint:* You do not need to complete a DC analysis - just the AC analysis.

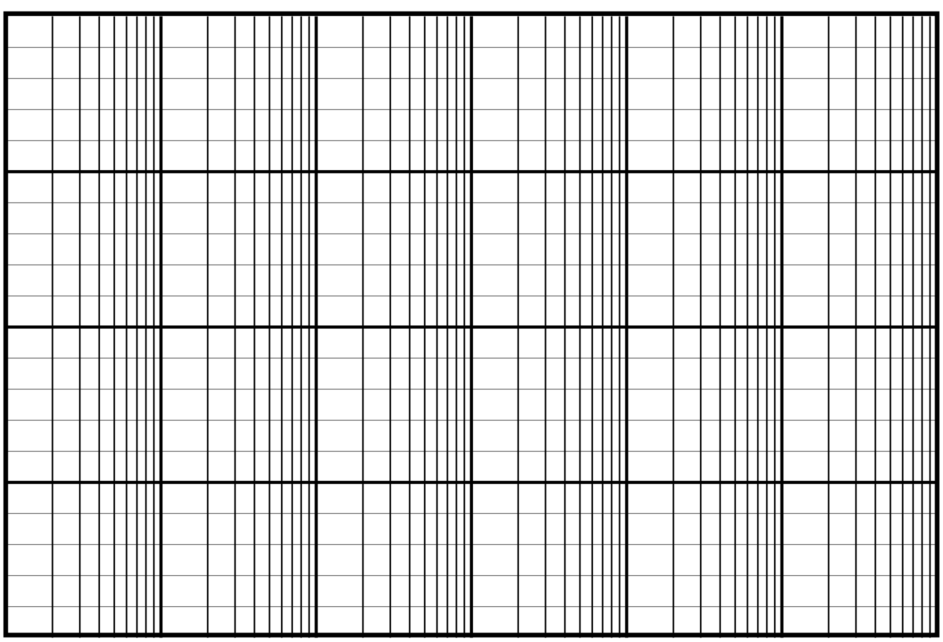
(b) (10 points) Find the expression for the *midband gain, A,* with the load and source removed. The solution should be expressed in terms of resistors, β, r_π.

(c) (10 points) Find expression for the input resistance, R_{in}, for the *midband*.
Hint: attach the load.

(d) (10 points) Find expression for the output resistance, R_{out}, for the *midband.*
Hint: attach the source.

(e) (10 points) Draw the equivalent circuit of the amplifier using the parameters identified above (R_{in}, R_{out}, and A) with the source and load attached. Find expressions for resistances R_{C1} and R_{C2} as seen by capacitors C_1 and C_2, respectively.

(f) (10 points) Assuming R_{in} = 5k, R_{out} = 5k, R_S = 5k, R_L= 5k, C_1 = 10nF, C_2 = 0.1uF, and midband gain A=400, plot the Bode plot of the transfer function for the circuit. Make sure to label the axes.



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End of Exam